Living Schematic → production · step 1/3: netlist batch + coverage v0.99.0

The 0.91 PoC inferred a netlist per page on demand. Now we precompute the whole corpus once, cache it, and measure coverage.

documents (index)

read id + path from viewer.db (read-only, R1). One task per document -> the PDF is opened once per worker.

scan pages

schem_overlay.schem_paths(page) -> has_vector? enough segments? MuPDF get_drawings; skip raster scans fast.

infer netlist

schemgraph.graph_from_paths: snap nodes, split T-junctions, union-find nets, attach refs, confidence score.

cache + coverage

-> index/schemcache/<doc>_<pg>.json (same cache /api/schemgraph serves) -> index/schemgraph_coverage.tsv resumable via schemgraph_done.txt

parallel: one worker per doc, cpu_count-1 (cap 12). BUILD-SCHEMGRAPH.bat (full) · VERIFY-SCHEMGRAPH.bat (--limit slice).

WHY THIS MATTERS

Instant Flow

The ▶ Flow overlay reads a precomputed JSON instead of inferring on open — no per-open lag.

Coverage truth

The TSV says exactly which pages have a usable netlist (nets/edges/confidence) — a map for steps 2 & 3.

Feeds review

Pages with wires-but-0-components (outlined label text) are the queue for step 2, the junction/label review.

Feeds Circuit Lab

High-confidence netlists are the candidates to push into the live MNA simulator (step 3).

VERIFIED (host-side, real corpus)

- VERIFY-SCHEMGRAPH.bat (--limit 200): the batch runs and cached 4,743 schematic-page netlists (e.g. 287 segments / 11 nets / conf 0.80).
- Cache is the exact format /api/schemgraph serves; coverage TSV + resumable done-file are plain text.
- Known: many pages report 0 components (CAD-exported sheets outline label text) -> the input to step 2 (review queue).
- Additive & rollbackable (R1): delete build_schemgraph.py + the bats + the two sidecar files.